

ALVIN VINCENT OSWALD REBA

Backend / AI Platform Engineer Intern

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Summary

Backend-focused software engineer with production-grade experience building systems using **Golang, Clean Architecture, and modern DevOps practices**. Experienced in designing secure multi-tenant APIs, IoT data pipelines, and scalable monitoring platforms using PostgreSQL, InfluxDB, and Redis. Comfortable working across the full backend stack—from system design and database modeling to CI/CD and zero-downtime deployment. Familiar with integrating **LLM-based systems and AI workflows**, with growing experience in building AI-powered services. Seeking an Engineering Intern role to contribute to robust AI-enabled products at scale.

Education

Sepuluh Nopember Institute of Technology (ITS)

Surabaya, Indonesia

Bachelor of Science in Informatics

Expected Graduation: 2026

- Relevant Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Software Engineering, Machine Learning

Technical Skills

Languages: Go (Golang), SQL, Python (basic)

Backend & Architecture: Clean Architecture, RESTful API Design, Gin Framework, Middleware Development, RBAC, Multi-tenant Systems, TDD

Databases: PostgreSQL (schema design, indexing, optimization), InfluxDB (time-series), Redis (caching, sessions)

DevOps & Infrastructure: Docker, Docker Compose, GitLab CI/CD, Zero-downtime Deployment, Cloudflare Tunnel, Trivy, gosec

Security: JWT Authentication, SSO Integration, Role-Based Access Control (RBAC), Secure API Design

AI / LLM: LLM API Integration, Prompt Engineering, Tool/Function Calling, AI Workflow Orchestration, Vector Search

Tools: Git, GitLab, Make/Makefile, golangci-lint, Swagger/OpenAPI

Experience

Independent Backend Engineer

2023 – Present

IoT Monitoring Systems

Remote

- Designed and developed multiple production-style backend systems independently from scratch to deployment
- Delivered secure, scalable RESTful APIs supporting real-time sensor data ingestion and visualization
- Applied software engineering best practices: CI/CD pipelines, automated testing, linting, and security scanning
- Collaborated with frontend developers to integrate APIs into dashboards and internal monitoring tools

Projects

LecSens – Water Sensor Monitoring Backend | *Golang, Gin, PostgreSQL, InfluxDB, Redis, Docker*

2024

- Designed and built a complete backend platform for water sensor monitoring as sole developer
- Implemented Clean Architecture to separate domain, application, and infrastructure layers for maintainability
- Built secure REST APIs with JWT authentication and role-based access control (RBAC)
- Designed PostgreSQL schema for operational data and InfluxDB for high-frequency time-series sensor readings
- Implemented caching and session management using Redis, improving response times by 40%
- Deployed using Docker with CI/CD automation, integrated security scanning (Trivy, gosec)

POME Monitoring System – Industrial Sensor Backend | *Golang, PostgreSQL, InfluxDB, Redis, Docker*

2025

- Built microservice-inspired backend for monitoring Palm Oil Mill Effluent (POME) sensors
- Integrated existing ML prediction model into backend via API for downstream analytics and dashboards
- Designed semi real-time data ingestion pipeline with time-series processing using InfluxDB
- Implemented multi-tenant architecture to securely isolate data between organizations
- Developed comprehensive API documentation and optimized database queries for high-frequency sensor data

Additional Information

Strengths: Strong problem-solving and system design mindset, comfortable in ambiguous startup environments

Work Style: End-to-end feature ownership (design → implementation → deployment)

Current Focus: Actively learning AI platform engineering and LLM-based system design

Languages: Indonesian (Native), English (Professional Working Proficiency)